INTERVIEWEE: Raymond F. Boyer

INTERVIEWER: James J. Bohning

LOCATION: Michigan Molecular Institute, Midland, Michigan

DATE: 14 January 1986

BOHNING: Dr. Boyer, you were born on February 6, 1910, in Denver, Colorado. Can you tell me something about your parents?

BOYER: Well, my father grew up on a farm in Mount Hope, Ohio. He wandered around a lot, finally ending up in Denver, Colorado, where he met my mother. She was born in Sioux City, Iowa and had gone to Denver as the metropolis of that area. So that's where they met. Several years after I was born, my father decided to move back to Ohio because his mother had died and his father was getting on in years and was all alone. They came back and first went to the village of Beach City, Ohio. They were there for several years, and then moved about twenty miles away to Millersburg, Ohio. That was the county seat of Holmes County, population around 2000 to 2500. There's where I had my schooling from grades one through nine.

BOHNING: Were there any particular teachers in Millersburg at that time that influenced you?

BOYER: I'd say the teachers by and large were excellent. They tended to be. Perhaps the eighth grade teacher left the most lasting impression, partly because she was a tyrant, but she was extremely good. She was Mrs. Carrie Marvin. My mother bought a set of books--The Book of Knowledge--which I read incessantly (1). They had a major influence on my career.

BOHNING: Did you go to high school in Millersburg?

BOYER: Only junior high, the ninth grade. My father was an entrepreneur. I think that is the best description for him. In Millersburg he operated a dry cleaning and laundry establishment, and for many years it was the only one in town. Then, competition moved in with some newer equipment. He decided that Millersburg did not need two such places and so he simply got out. For several years, he leased land for oil, using the old witch hazel stick to prospect for oil.

BOHNING: Was that in the Millersburg area?

BOYER: Yes. Well, it was all over Ohio. Ohio was a hot state then for oil potential and so that's what he did. That didn't last very long, and then he took a temporary position in Cleveland, Ohio, operating a confectionery stand with popcorn, peanuts, salted nuts, and so on. He didn't like being away from home, living in Millersburg, and working in Cleveland, so he decided to move to Canton, Ohio, which then had a population of about a quarter million. Again, he had a position then at a park concession, with popcorn, peanuts, and soft drinks. From there, he went into business for himself in a new market which was opening up in Canton. I worked after school in that market of his for about three years--after school, Saturdays, nights, and so on. That led to a tug of war between him and my mother. She wanted me to go to college. He wanted me to stay and inherit his business, so to speak. Well, she won. But I went to high school in Canton, Ohio. It had one senior high school for a town of a quarter million. It had about four thousand students. That was grades ten through twelve. Because I did very well in the first year, I got put in a class for outstanding students with special teachers--the best.

BOHNING: What influenced you to move into physics?

BOYER: Since I was about ten years old I had an interest in electricity. I always had a shop in my house. It was mostly full of junk, but important to me. I think, as far as I know, this was probably the Thomas Edison influence in Ohio. Also, there was a magazine by Hugo Gernsback called The Electrical Experimenter (2). Those were the influences that were probably critical to my career from about ten on. For some reason I never went into radio. I had friends who did. But no, it was Thomas Edison, that's all I can imagine. Now, I don't know if you're interested in the choice of Case?

BOHNING: Yes, absolutely. But before we get to Case, were there any teachers in your high school in Canton that you remember, in terms of the chemistry or the physics or the math that you may have taken, that influenced you?

BOYER: Yes. There was a Miss Heinreichs in chemistry. There was a Miss Lutz in Latin, and a Mr. Metzger in physics. There was a superb math teacher whose name I can't remember, but he was teaching advanced math then.

BOHNING: Did you have any leanings towards physics over chemistry?

BOYER: No. I'll get to that. I missed being valedictorian of the class by a bad grade in physics. [laughter] I just goofed on one exam. But anyway, I think the choice of Case is rather crucial because Case led to Dow. A high school friend of mine and I had looked around for colleges, and we traveled. I wanted electrical engineering--no question about it--and he wanted mechanical. So we made the rounds from Carnegie Tech to Ohio State to Purdue, Notre Dame, University of Chicago, and we couldn't make a decision. About a month before college was to start, in the fall of 1929, we were visiting at a friend's house, and the mother said, "Why don't you go to Case? There's a famous man there who works in musical sound and has written a book on the flute (3). His name is Dr. Miller." Well, we grabbed it; without ever having seen the campus which was only sixty miles away, we enrolled.

BOHNING: So you visited all the other schools! This Dr. Miller, was he in physics?

BOYER: He was Dayton C. Miller, who was in acoustics. The musical sound was a hobby. He was doing ether drift experiments, and was a wonderful lecturer. I was still committed to electrical engineering when I went to Case and I indicated that choice on my enrollment form. At a fraternity rushing situation, I met a man who was working in radio propagation at the Bureau of Standards. When I told him my interests, he said, "You don't want to be an electrical engineer, you want to be a physicist." And that was it.

BOHNING: Were there any specific faculty at Case that influenced you?

BOYER: Well, yes. Because in a placement exam I was in the top ten, it turned out that we got some good teachers that way. One of them was the professor of astronomy. His name was Jason J. Nassau. He taught freshman math. I think he was the first one to recognize any potential in me, and he remained a lifelong friend, advisor and mentor.

BOHNING: What about physics?

BOYER: My father died at the end of the freshman year, during exam week. Sometime after that, I had a chance encounter with Professor Nassau and told him that I didn't think I could go back to school because of a lack of funds. This was in the spring of 1930 and the Depression was on. He told me that there was an opening at the school observatory for a student custodian and he thought he could get that for me. Well, he did. So, I moved into that observatory. I had a shared free room, a kitchen with

cooking privileges, bath, and so on. That really was what allowed me to go to school. My roommate and I swept and scrubbed floors and cleaned toilets as the exchange for staying there.

I enrolled then in physics at the start of the sophomore year and stayed with it throughout. Nassau was a proselytizer and he persuaded me to do a thesis in astronomy. I had been familiar with all the equipment. I had assisted at public nights and I knew how to run the telescope. I knew the stars, and the whole business just by living there. He was certainly the dominant figure. He was nominally a member of the physics department. But the rest of the staff there, well, they weren't very good. They weren't outstanding. Miller was approaching retirement. The editor of the Handbook of Chemistry and Physics(4), [Charles D.] Hodgman, was on the staff. He let the students teach themselves. We'd get assigned a chapter in a book on optics and we gave the lectures. He sat back and did nothing. There were six of us enrolled in that physics course.

[Christian] Nusbaum was in x-ray crystallography. He was quite good, but he never amounted to anything in the big scene. There was a man named Clarence William Wallace who was simply drawing his pay. He taught, but indifferently. There was [John G.] Albright who was a good teacher, but his chief claim to fame was that he discovered that spider webs made very good cross hairs in microscopes. He had a business to supply spider webs. There were some good people. Of course, you took courses all over the campus in different disciplines. The math was generally very good. So there were a few good teachers around, enough that I had a decent education. Later, during my master's degree work, I had a course in physical metallurgy with A. A. Bates, who achieved national fame. [Robert S.] Shankland, a physicist, was very good but was away working with [Arthur H.] Compton on a Ph.D. degree while I was there.

BOHNING: How many chemistry courses did you take?

BOYER: Three: general; analytical laboratory; and physical chemistry.

BOHNING: Do you recall who you had?

BOYER: I don't remember analytical. General chemistry was taught by William Reed Veazey. Have you ever come across him?

BOHNING: Yes. I'm sure we're going to mention his name some more before we're finished, because he was already, at the time you were there, working at Dow in the summers and consulting for Dow.

BOYER: Yes, and then later he moved there permanently. Veazey was on Dow's Board of Directors and director of research for Dow starting prior to the death of Willard Dow in 1949. I worked for him at Dow later; I was an assistant to him for three years. The physical chemistry class was taught by Norbert Lange. He was the co-editor of the Handbook of Chemistry and Physics (4). I think that's right, Hodgman and Lange. He was excellent as a teacher.

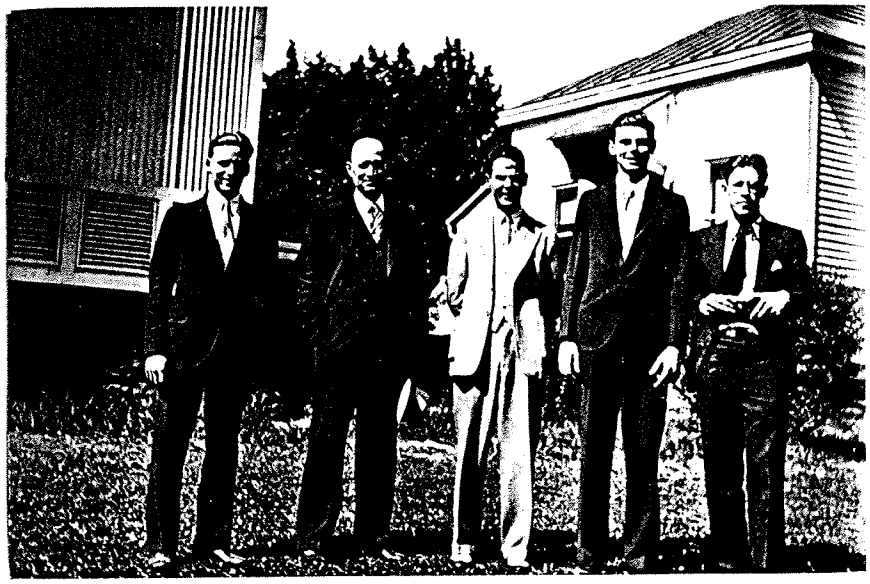
BOHNING: You took general chemistry and physical chemistry. No organic?

BOYER: No organic. I never had a course in organic chemistry. Somewhere along the line there was a lab in analytical chemistry,and that was the extent of it.

I first learned the terms "polymer" and "polymerization" at the observatory. A Case graduate, Frank Herzog, lived in Cleveland and worked in Akron (Goodrich). He was an amateur photographer and used the fully equipped observatory darkroom mostly on weekends. On one occasion, he brought a 2-3 liter bottle of vinyl acetate to this darkroom and left it there. It had an Eastman Kodak label. This liquid gradually thickened. I knew because I cleaned the darkroom. My curiosity made me tip the bottle and learn that this liquid gradually became thicker. Then I questioned Frank about my observations. The bottle originally contained a monomer, vinyl acetate, which he was testing in photographic chemistry. At room temperature it gradually polymerized and thickened the monomer. I can still "see" in "my mind's eye" that large bottle of vinyl acetate. The exact meaning of this finally became clear to me after several years at Dow Chemical. Herzog later was inventor (or co-inventor) of the tubeless tire.

BOHNING: You said you did a thesis. That was for the bachelor's degree in astronomy. Then you continued on for an M.S. degree?

BOYER: Well, when I tried to get a job in the spring of 1933, I went around to the industrial centers, and they wanted to know what my thesis was. When I said astronomy, that was the end. They made it very clear that they couldn't possibly want an astronomer. [laughter] I didn't really see how I could possibly go back, but then some teaching assistantships opened up in the fall of 1933. I think it was Nassau who got me one of them. Now I was in the physics department at Case, and I was a laboratory instructor. This was a two-year arrangement, fifty dollars a month, part of which went home to support my mother. That got to be pretty exciting. But I decided that I wasn't going to do any more astronomy even though Nassau wanted me to. So I signed up with Nusbaum, and the master's thesis was on the x-ray



crystallography of steel at high temperature in the region of the so-called alpha transition.

BOHNING: Was that done in conjunction with any of the steel companies in Cleveland?

BOYER: No. I don't know whether Nusbaum consulted with any of them or not. It's a good possibility, but I wasn't aware of that. And the joke of it was, we had a persistent problem in getting the photographs and Nusbaum was of no help. On the day I graduated, I realized what the problem was and I solved it. It was so damned simple that it was ironic and I started turning out decent x-ray spectrographs. The red hot iron specimen darkened the photographic film! That made the thesis.

BOHNING: What happened to your friend who went with you to Case, and were there any other students with you at the time that you recall?

BOYER: Well, my friend went through Case and had an indifferent four years in mechanical engineering, got an indifferent job somewhere in Cleveland and did nothing with his career. We roomed together freshman year and lunched together sophomore year on sandwiches I made at the observatory.

BOHNING: Were there any other student colleagues?

BOYER: The original five of us in the physics department were a rather remarkable group for the most part. There was Briggs Hazelhurst Napier--he was a descendant of the two men who invented logarithms, [John] Napier and [Henry] Briggs, and was quite a brain. He never made anything out of his life, did some teaching, and had a mental breakdown--nothing. There was a man named Robert A. Herrington, who ended up at Goodrich. He was very brilliant. He never did anything with his career at Goodrich. He was useful. For instance, he had memorized a four-place logarithm table. [laughter] But that was the kind of person he was. These guys were so sharp that they used to make me feel like an ignoramus. Another man was named Irving Prettyman. He ended up at Firestone, and he retired from there. I'd say he didn't distinguish himself at all. He just did good work on polymer and rubber physics.

Our original group of five was joined by a sixth student for the senior year--a William Nels Simons from New Orleans. We commonly designated him as the one who speaks Southern drawl with a Harvard accent (or perhaps vice versa). He came to class impeccably dressed--full business suit, clean shirt each day, and tie. I, for example, looked like a tramp. [see following page]